

Assembling Functions

David Puelz

The University of Austin

Outline

1. **Linear Combination** – Adding scaled functions
2. **Function Composition** – Functions of functions
3. **Function Multiplication** – Multiplying function outputs
4. **Piecewise Functions** – Different formulas on different domains
5. **Examples** – Step-by-step assembly

1 Linear Combination

1.1 Definition

Given functions $g_1(x), g_2(x), \dots, g_n(x)$ and coefficients $c_1, c_2, \dots, c_n$:

$$f(x) = c_1 g_1(x) + c_2 g_2(x) + \dots + c_n g_n(x)$$

The coefficients c_i are parameters that scale each function before adding.

1.2 Example: Polynomials

Polynomials are linear combinations of power functions.

Example: $f(x) = 3x^2 + 5x - 2$

Step 1: Identify component functions

$$g_0(x) = 1 \quad (\text{constant})$$

$$g_1(x) = x \quad (\text{identity})$$

$$g_2(x) = x^2 \quad (\text{quadratic})$$

Step 2: Write as linear combination

$$f(x) = 3 \cdot g_2(x) + 5 \cdot g_1(x) + (-2) \cdot g_0(x)$$

Step 3: Verify

$$\begin{aligned} f(x) &= 3 \cdot x^2 + 5 \cdot x + (-2) \cdot 1 \\ &= 3x^2 + 5x - 2 \quad \checkmark \end{aligned}$$

1.3 Example: Sinusoid with Baseline

Model: $f(t) = A \sin(\omega t + \phi) + B$

Step 1: Identify components

$$g_1(t) = 1 \quad (\text{constant})$$

$$g_2(t) = \sin(\omega t + \phi) \quad (\text{scaled sinusoid})$$

Step 2: Write as linear combination

$$f(t) = A \cdot g_2(t) + B \cdot g_1(t) = A \sin(\omega t + \phi) + B$$

Step 3: Parameter interpretation

- A : Amplitude (coefficient of sinusoid)
- B : Baseline (coefficient of constant)
- ω, ϕ : Inside the function (not coefficients)

1.4 Example: Two Peaks (HW2 Problem 2)

Model: $G(t) = B + A_1 h\left(\frac{t-\mu_1}{s_1}\right) + A_2 h\left(\frac{t-\mu_2}{s_2}\right)$

where $h(x) = e^{-x^2}$ is the hump function.

Step 1: Identify components

$$g_0(t) = 1 \quad (\text{baseline})$$

$$g_1(t) = h\left(\frac{t-\mu_1}{s_1}\right) \quad (\text{morning peak})$$

$$g_2(t) = h\left(\frac{t-\mu_2}{s_2}\right) \quad (\text{evening peak})$$

Step 2: Write as linear combination

$$G(t) = B \cdot g_0(t) + A_1 \cdot g_1(t) + A_2 \cdot g_2(t)$$

Step 3: Parameter roles

- B : Baseline occupancy (coefficient of constant)
- A_1 : Height of morning peak (coefficient of first hump)
- A_2 : Height of evening peak (coefficient of second hump)
- μ_1, μ_2 : Peak locations (inside hump functions)
- s_1, s_2 : Peak widths (inside hump functions)

2 Function Composition

2.1 Definition

Given functions $g(x)$ and $h(x)$, composition means:

$$f(x) = g(h(x))$$

The output of h becomes the input of g .

Order matters: $g(h(x)) \neq h(g(x))$ in general.

2.2 Example: Scaled Sinusoid

Model: $f(t) = \sin(2t + 3)$

Step 1: Identify inner and outer functions

$$h(t) = 2t + 3 \quad (\text{inner: linear function})$$

$$g(x) = \sin(x) \quad (\text{outer: sinusoid})$$

Step 2: Compose

$$f(t) = g(h(t)) = \sin(2t + 3)$$

Step 3: Evaluate at specific point

$$f(0) = \sin(2(0) + 3) = \sin(3) \approx 0.141$$

$$f(\pi) = \sin(2\pi + 3) = \sin(3) \approx 0.141$$

2.3 Example: Exponential of Linear Function

Model: $f(t) = e^{2t-1}$

Step 1: Identify inner and outer

$$h(t) = 2t - 1 \quad (\text{inner: linear})$$

$$g(x) = e^x \quad (\text{outer: exponential})$$

Step 2: Compose

$$f(t) = g(h(t)) = e^{2t-1}$$

Step 3: Verify properties

$$f(0) = e^{2(0)-1} = e^{-1} \approx 0.368$$

$$f(1) = e^{2(1)-1} = e^1 \approx 2.718$$

2.4 Example: Nested Composition

Model: $f(x) = \log(x^2 + 1)$

Step 1: Identify all functions

$$h_1(x) = x^2 \quad (\text{innermost})$$

$$h_2(x) = x + 1 \quad (\text{middle})$$

$$g(x) = \log(x) \quad (\text{outermost})$$

Step 2: Compose step-by-step

$$\text{Step 2a: } h_1(x) = x^2$$

$$\text{Step 2b: } h_2(h_1(x)) = x^2 + 1$$

$$\text{Step 2c: } f(x) = g(h_2(h_1(x))) = \log(x^2 + 1)$$

Step 3: Domain considerations

- $h_1(x) = x^2$: Domain is all real numbers
- $h_2(x) = x + 1$: Domain is all real numbers
- $g(x) = \log(x)$: Domain is $x > 0$
- Therefore: $f(x)$ requires $x^2 + 1 > 0$, which is always true
- Final domain: All real numbers

3 Function Multiplication

3.1 Definition

Given functions $f(x)$ and $g(x)$, multiply their outputs:

$$h(x) = f(x) \cdot g(x)$$

$$\text{Commutative: } f(x) \cdot g(x) = g(x) \cdot f(x)$$

3.2 Example: Damped Oscillation

Model: $y(t) = e^{-t} \sin(t)$

Step 1: Identify component functions

$$f(t) = e^{-t} \quad (\text{exponential decay})$$

$$g(t) = \sin(t) \quad (\text{sinusoid})$$

Step 2: Multiply

$$y(t) = f(t) \cdot g(t) = e^{-t} \sin(t)$$

Step 3: Behavior analysis

- At $t = 0$: $y(0) = e^0 \sin(0) = 1 \cdot 0 = 0$
- As $t \rightarrow \infty$: $e^{-t} \rightarrow 0$, so $y(t) \rightarrow 0$ (oscillation dies out)
- The exponential envelope e^{-t} modulates the amplitude of $\sin(t)$

3.3 Example: Wave Packet

Model: $f(x) = e^{-x^2} \cos(5x)$

Step 1: Identify components

$$\begin{aligned} g_1(x) &= e^{-x^2} \quad (\text{Gaussian envelope}) \\ g_2(x) &= \cos(5x) \quad (\text{oscillation}) \end{aligned}$$

Step 2: Multiply

$$f(x) = g_1(x) \cdot g_2(x) = e^{-x^2} \cos(5x)$$

Step 3: Localization

- The Gaussian e^{-x^2} is large near $x = 0$ and small far away
- The cosine $\cos(5x)$ oscillates rapidly
- Product: Localized oscillation (wave packet)¹

4 Piecewise Functions

4.1 Definition

A function defined by different formulas on different parts of its domain:

$$f(x) = \begin{cases} f_1(x) & \text{if condition 1} \\ f_2(x) & \text{if condition 2} \\ \vdots & \end{cases}$$

4.2 Example: Absolute Value

Definition: $|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$

¹ A **wave packet** is a localized burst of oscillation—e.g., a sinusoid multiplied by a bell-shaped (Gaussian) envelope so the wave is concentrated in a region and dies off elsewhere. Applied contexts: quantum mechanics (localized particle states), optics (pulsed laser beams), acoustics (sound bursts), and radar/sonar (short pulses used for ranging).

Step 1: Identify pieces

Piece 1: $f_1(x) = x$ for $x \geq 0$

Piece 2: $f_2(x) = -x$ for $x < 0$

Step 2: Verify at boundary

At $x = 0$: $f_1(0) = 0$

$f_2(0) = -0 = 0$ ✓ (continuous)

Step 3: Evaluate examples

$|x|$ at $x = 3$: $3 \geq 0 \Rightarrow |3| = 3$

$|x|$ at $x = -3$: $-3 < 0 \Rightarrow |-3| = -(-3) = 3$

4.3 Example: Coffee Cooling with Milk (HW2 Problem 1g)

Setup: Coffee cools according to Newton's law, then milk is added at $t = 10$ minutes.

Step 1: Original model

$$T(t) = T_{\text{room}} + (T_0 - T_{\text{room}})e^{-kt}$$

Given: $T_{\text{room}} = 22$, $T_0 = 95$, $T(10) = 69$.

Step 2: Solve for k

$$69 = 22 + (95 - 22)e^{-10k}$$

$$69 - 22 = 73e^{-10k}$$

$$47 = 73e^{-10k}$$

$$\frac{47}{73} = e^{-10k}$$

$$\ln\left(\frac{47}{73}\right) = -10k$$

$$k = -\frac{1}{10} \ln\left(\frac{47}{73}\right) \approx 0.0434 \text{ min}^{-1}$$

Step 3: Model before milk ($t < 10$)

$$T(t) = 22 + 73e^{-0.0434t}, \quad t \in [0, 10)$$

Step 4: Temperature at $t = 10$ (just before milk)

$$T(10) = 22 + 73e^{-0.0434(10)} = 22 + 73e^{-0.434} \approx 22 + 47.0 = 69.0^\circ\text{C}$$

Step 5: Temperature after adding milk Milk drops temperature by 3°C instantly:

$$T(10^+) = 69 - 3 = 66^\circ\text{C}$$

Step 6: New initial condition for $t \geq 10$ After milk, the coffee continues cooling with:

- New initial temperature: $\tilde{T}_0 = 66^\circ\text{C}$
- Same room temperature: $T_{\text{room}} = 22^\circ\text{C}$
- Same cooling rate: $k = 0.0434 \text{ min}^{-1}$

Step 7: Model after milk ($t \geq 10$) We need to shift time so that at the shifted time $t' = 0$, we have $\tilde{T}_0 = 66$.

For $t \geq 10$, let $t' = t - 10$ (time since adding milk):

$$\tilde{T}(t) = 22 + (66 - 22)e^{-k(t-10)} = 22 + 44e^{-0.0434(t-10)}, \quad t \geq 10$$

Step 8: Verify continuity

$$\begin{aligned} \text{At } t = 10: \quad \tilde{T}(10) &= 22 + 44e^{-0.0434(10-10)} \\ &= 22 + 44e^0 = 22 + 44 = 66^\circ\text{C} \quad \checkmark \end{aligned}$$

Step 9: Complete piecewise function

$$\tilde{T}(t) = \begin{cases} 22 + 73e^{-0.0434t} & \text{if } 0 \leq t < 10 \\ 22 + 44e^{-0.0434(t-10)} & \text{if } t \geq 10 \end{cases}$$

Step 10: Find time to reach 60°C Since $60 > 22$ (above room temperature), we need to check which piece applies.

Case 1: $t < 10$ (before milk)

$$\begin{aligned} 60 &= 22 + 73e^{-0.0434t} \\ 38 &= 73e^{-0.0434t} \\ \frac{38}{73} &= e^{-0.0434t} \\ \ln\left(\frac{38}{73}\right) &= -0.0434t \\ t &= -\frac{1}{0.0434} \ln\left(\frac{38}{73}\right) \approx 12.8 \text{ minutes} \end{aligned}$$

But $12.8 > 10$, so this solution is invalid (we're in the second piece).

Case 2: $t \geq 10$ (after milk)

$$60 = 22 + 44e^{-0.0434(t-10)}$$

$$38 = 44e^{-0.0434(t-10)}$$

$$\frac{38}{44} = e^{-0.0434(t-10)}$$

$$\ln\left(\frac{38}{44}\right) = -0.0434(t-10)$$

$$t-10 = -\frac{1}{0.0434} \ln\left(\frac{38}{44}\right)$$

$$t = 10 - \frac{1}{0.0434} \ln\left(\frac{38}{44}\right) \approx 10 + 3.4 = 13.4 \text{ minutes}$$

Answer: Coffee reaches 60°C at $t \approx 13.4$ minutes.

5 Combining Methods

5.1 Example: Complex Assembly

Model: $f(x) = 2e^{-x^2} \sin(3x) + 1$

Step 1: Identify all operations

1. **Composition:** $\sin(3x)$ (inner: $3x$, outer: $\sin$)
2. **Composition:** e^{-x^2} (inner: $-x^2$, outer: e^x)
3. **Multiplication:** $e^{-x^2} \cdot \sin(3x)$
4. **Linear combination:** $2 \cdot [e^{-x^2} \sin(3x)] + 1 \cdot 1$

Step 2: Build step-by-step

Step 2a: $h_1(x) = 3x$

Step 2b: $g_1(x) = \sin(h_1(x)) = \sin(3x)$

Step 2c: $h_2(x) = -x^2$

Step 2d: $g_2(x) = e^{h_2(x)} = e^{-x^2}$

Step 2e: $m(x) = g_2(x) \cdot g_1(x) = e^{-x^2} \sin(3x)$

Step 2f: $f(x) = 2 \cdot m(x) + 1 = 2e^{-x^2} \sin(3x) + 1$

Step 3: Evaluate at $x = 0$

$$\begin{aligned} f(0) &= 2e^{-0^2} \sin(3 \cdot 0) + 1 \\ &= 2e^0 \sin(0) + 1 \\ &= 2(1)(0) + 1 = 1 \end{aligned}$$

6 Summary

6.1 Four Methods for Assembling Functions

1. **Linear Combination:** $f(x) = c_1g_1(x) + c_2g_2(x) + \dots$
 - Scale each function by a coefficient, then add
 - Examples: Polynomials, Fourier series, regression models
2. **Composition:** $f(x) = g(h(x))$
 - Output of inner function becomes input of outer function
 - Order matters: $g(h(x)) \neq h(g(x))$ in general
 - Examples: $\sin(2t + 3)$, e^{x^2} , $\log(x^2 + 1)$
3. **Multiplication:** $f(x) = g(x) \cdot h(x)$
 - Multiply outputs of two functions
 - Commutative: $g(x)h(x) = h(x)g(x)$
 - Examples: Damped oscillation, wave packets
4. **Piecewise:** Different formulas on different domains
 - Use conditions to select which formula applies
 - Examples: Absolute value, coffee with milk

6.2 Key Points for HW2

- **Problem 1g:** Piecewise function for coffee cooling with milk addition
- **Problem 2:** Linear combination of hump functions for two peaks
 - $G(t) = B + A_1h\left(\frac{t-\mu_1}{s_1}\right) + A_2h\left(\frac{t-\mu_2}{s_2}\right)$
 - Parameters: B (baseline), A_1, A_2 (peak heights), μ_1, μ_2 (locations), s_1, s_2 (widths)
- **Problem 4:** Multiple inputs (linear combination of inputs, not functions)
 - $\widehat{\text{rating}} = \beta_0 + \beta_1r + \beta_2v$
 - This is a function of two variables: $f(r, v)$